SYNOPSIS

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Project Title: Prediction of Crop Yield Based-on Soil Parameters using Machine Learning Algorithms

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Abstract

Smart Agriculture helps farmers choose crops based on current soil and weather conditions. This project builds a crop recommendation system that reads key soil parameters—nitrogen (N), phosphorus (P), potassium (K), and pH—using a multi-parameter soil sensor. It also fetches temperature, humidity, and rainfall from online APIs.

An ESP32 collects the sensor readings and sends them to a Raspberry Pi. On the Pi, an ensemble model trained with boosting combines the soil and weather inputs and predicts the best crop for those conditions. The system monitors the inputs continuously and shows the recommendation in a simple interface.

We tested the system across multiple environmental scenarios and it produced steady outputs. The result is a practical tool to improve Agriculture yield while reducing wasted water and fertilizer.

Specific Contribution

- Built the Data gathering setup with soil sensors and an ESP32 to monitor readings in real time.
- Set up the ESP32-to-Raspberry Pi link and handled data transfer between hardware and the processing unit.
- Owned the end-to-end pipeline: collect data, run the model, and display the crop recommendation.

Specific Learning

- Learned how IoT-based Agriculture systems are put together, from sensors to dashboards.
- Understood how soil and weather readings influence crop recommendation in practice.
- Learned to use ensemble methods (including boosting) to improve prediction accuracy.

Technical Limitations & Ethical Challenges faced

- Sensor readings weren't consistent, so calibration was needed to keep results dependable.
- The system depended on a steady internet connection to pull weather/API data.
- Avoiding biased recommendations across varied farming conditions took extra care in data selection and testing.

Keywords: Smart Agriculture, Crop Recommendation, IoT, Machine Learning, Ensemble Models, Soil Sensors, Precision Farming

Mythili

Signature of the Student **Signature of Guide**

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